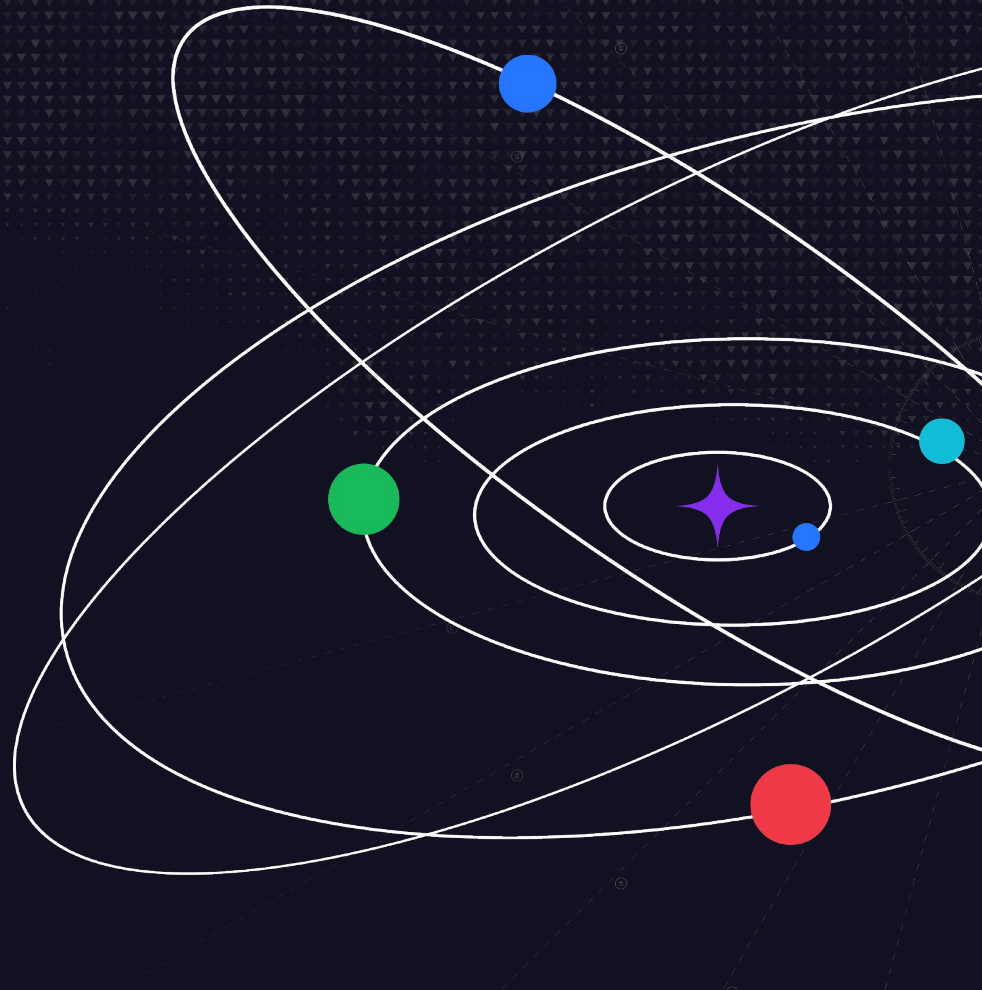


UX to manage Dags to Assets to Agents

The evolution of the Airflow UI

ASTRONOMER



What's Changed Since Last Summit

ASTRONOMER

UI plugin extension

Core integration

Connects your custom plugins with core Airflow

Real Object Passing: Passes actual instances

Theme Inheritance: Automatically inherits core styling

UI plugin extension

Core integration

Connects your custom plugins with core Airflow

Real Object Passing: Passes actual instances

Theme Inheritance: Automatically inherits core styling

AI-Driven dev & Proof

Accelerate development

AI Agent Rules: Ships with rules

HITL Review Plugin:
Human-In-The-Loop review capabilities (commonai provider).

UI plugin extension

Core integration

Connects your custom plugins with core Airflow

Real Object Passing: Passes actual instances

Theme Inheritance: Automatically inherits core styling

AI-Driven dev & Proof

Accelerate development

AI Agent Rules: Ships with rules

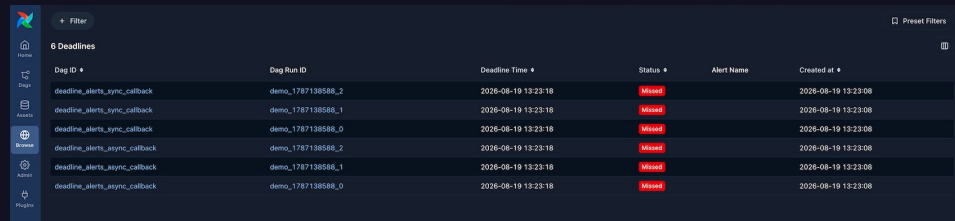
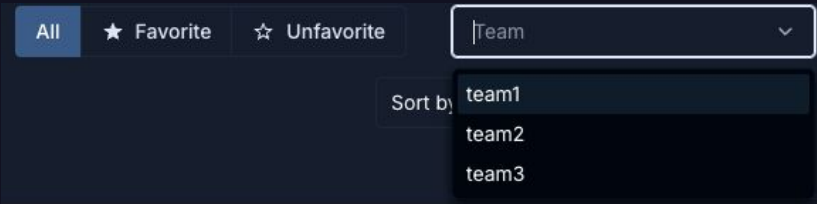
HITL Review Plugin:
Human-In-The-Loop review capabilities (commonai provider).

Expanded Targets

New destinations:

- asset
- base
- dag_overview
- task_overview

Multi team & Deadline Alerts



UI at scale

Pagination

Cursor based pagination

Heavy Endpoints: TI and Dag run lists

UI at scale

Pagination

Cursor based pagination

Heavy Endpoints: TI and Dag run lists

Grid Optimization

Engine Overhaul: fine-grained rendering strategies

Slick Views: virtualization and other UI techniques

UI at scale

Pagination

Cursor based pagination

Heavy Endpoints: TI and Dag run lists

Grid Optimization

Engine Overhaul: fine-grained rendering strategies

Slick Views: virtualization and other UI techniques

Graph Improvements

Decoupled Layouts: Leverage background ELK workers.

Fluid Experience: Fine-tuned default on large graphs.

Dag versioning tools

Trigger time

Bundle Selection: Pick the bundle version at trigger time to guarantee absolute execution consistency.

Historical runs

True Rendering: Historical runs render exactly as the DAG looked back then.

Precise Resolution: Correct task-group resolution

Dag versioning tools

Diagnostic

[Static Checker](#): Detect and locate dynamic fields

[Inflation Guide](#): Comprehensive guide

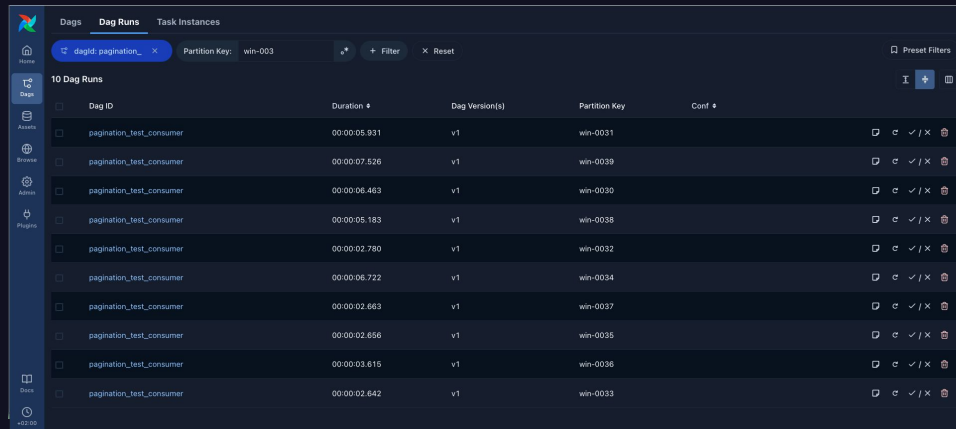
⚠ This DAG uses runtime-variable values in DAG construction. ⚠ It causes the DAG version to increase as values change on every DAG parse. Line 36, Col 6 Code: `DAG(dag_id=dag_id, default_args=default_args, description=f'dynamic_dag_id', schedule=timedelta(minutes=1), catchup=False, is_paused_upon_creation=False)` Issue: Don't use runtime-varying value as argument in DAG constructor Line 53, Col 8 Code: `PythonOperator(task_id=f'task_id_{rd.randint(1, 1000)}', python_callable=print_hello, dag=dag)` Issue: Don't use runtime-varying value as argument in TASK constructor

Partitions: data-aware runs (AIP-76)

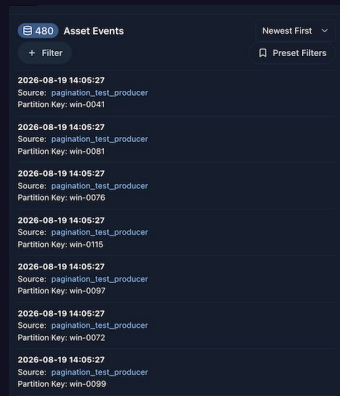
Runtime partitioning

Authoring: Emit partitioned asset events and combine events with partition mappers

Dependency Hold: Automatically hold downstream DAG runs until all required upstream partitions arrive safely.



Dag ID	Duration	Dag Version(s)	Partition Key	Conf
pagination_test_consumer	00:00:05.931	v1	win-0031	
pagination_test_consumer	00:00:07.526	v1	win-0039	
pagination_test_consumer	00:00:06.463	v1	win-0030	
pagination_test_consumer	00:00:05.183	v1	win-0038	
pagination_test_consumer	00:00:02.780	v1	win-0032	
pagination_test_consumer	00:00:06.722	v1	win-0034	
pagination_test_consumer	00:00:02.663	v1	win-0037	
pagination_test_consumer	00:00:02.656	v1	win-0035	
pagination_test_consumer	00:00:03.615	v1	win-0036	
pagination_test_consumer	00:00:02.642	v1	win-0033	



Source	Partition Key
pagination_test_producer	win-0041
pagination_test_producer	win-0081
pagination_test_producer	win-0076
pagination_test_producer	win-0115
pagination_test_producer	win-0097
pagination_test_producer	win-0072
pagination_test_producer	win-0099

Upgraded Human in the Loop UX

Required Actions Inbox

Review every pending approval and input across all Dags from a single cross-Dag queue. No more manually navigating between dag pages and task tabs.

🔔 7

Required Actions >

Dags

Dag Runs

Task Instances

Required Actions ⓘ

View all required actions

Pending required actions (3)

Dag ID	Dag Run	Map Index	Task ID	Created at
example_hit_operator	2026-08-20 07:37:25		wait_for_option	2026-08-20 07:37:27
example_hit_operator	2026-08-20 07:37:25		wait_for_multiple_options	2026-08-20 07:37:27
example_hit_operator	2026-08-20 07:37:25		wait_for_input	2026-08-20 07:37:27

Dag ID

example_hit_operator

Dag Run ID

manual__2026-08-20T11:37:25.934020+00:00

Map Index

-

Task ID

wait_for_multiple_options

Created at

2026-08-20 07:37:27 (3 minutes ago)

Try Number

1

Please choose option to proceed:

Options *

Select one or multiple values

Choose your options for this task instance

Respond

-04:00

User

Where the UI Has to Go Next

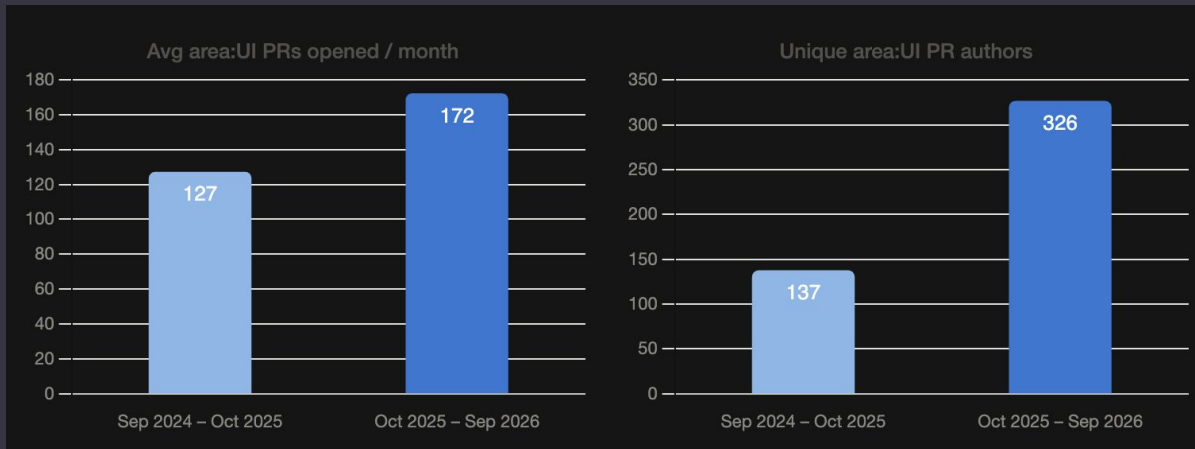
ASTRONOMER

Short Term

Stricter PR reviews

Enforce more code quality and UX quality

Double down on CI rules and agent tooling. Storybook tooling.



Short Term

Streamlining UX

Better handle unstructured data

Fewer clicks to logs

Less noise

More real time

More customizable

The screenshot displays the Apache Airflow web interface. The top navigation bar includes a search bar and a 'Trigger' button. The left sidebar contains navigation links for Home, Dags, Assets, Browse, and Admin. The main content area shows a task run for 'run_query [2]' with a success status. The 'Storage' tab is active, displaying a table of task state store entries. The 'XCom' tab is also visible, showing a key-value pair for 'return_value'.

Key	timestamp	Value
return_value	2026-08-27 15:51:37	<pre>{ "query": "SELECT\n \"Which version of Airflow do you currently use?\" AS label,\n COUNT(*) AS n\nPI\nsurvey\nWHERE \"Are you using AI/LLM (ChatGPT/Cursor/Claude etc) to assist you in writing Airflow co\n\" IN ('Always using AI', 'Often using AI', 'Sometimes using AI', 'Rarely using AI')\n AND \"Which\nversion of Airflow do you currently use?\" IS NOT NULL\nGROUP BY \"Which version of Airflow do you\ncurrently use?\"\nORDER BY n DESC;\", \"data\": [{ \"label\": \"2.8-2.11+\", \"n\": 1935 }, { \"label\": \"3.0-3.1\", \"n\": 1223 }] }</pre>

ASTRONOMER

Medium Term

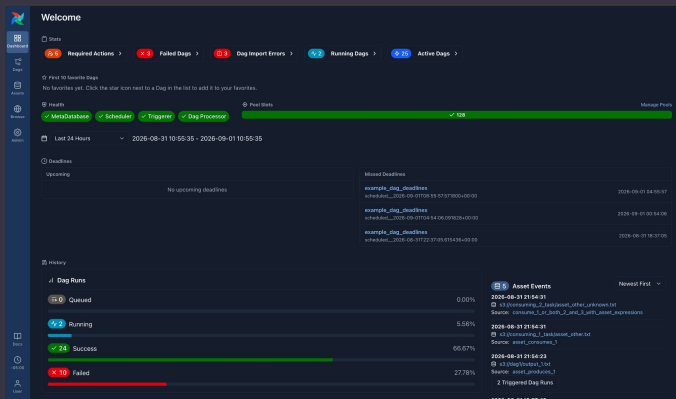
Overhauling

Dashboard and

Overview pages

ASTRONOMER

What is important information for your deployment, dag or task?



Medium Term

Expanding Our Plugin System

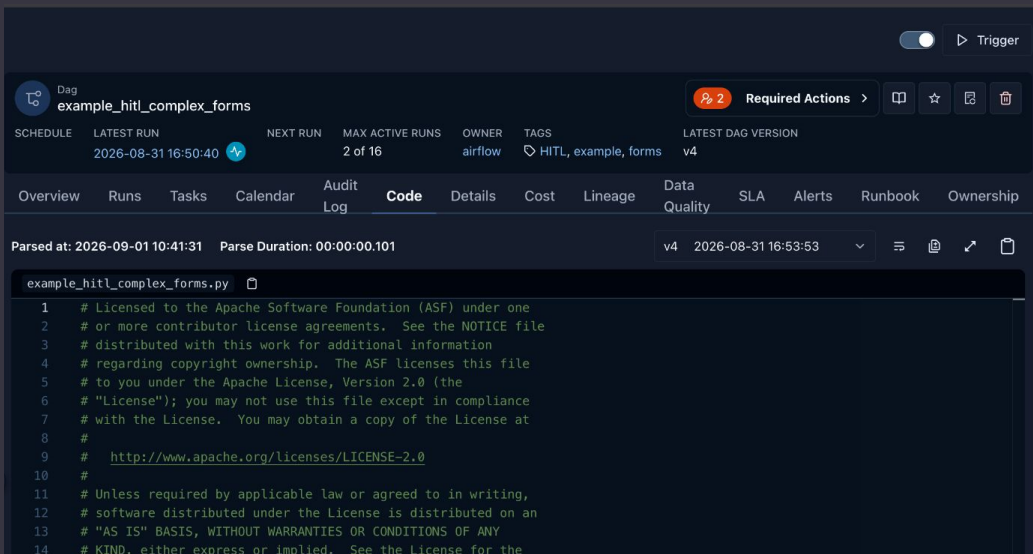
ASTRONOMER

Plugin Registry

Officially supported plugins for common providers

Expose more areas to plug into (dag actions, modals)

Shared UI code package



Medium Term

Filling in feature gap

ASTRONOMER

Pluggable Retry Policies

Asset Partition Mapping & Asset Watchers

Task & Asset State Store

The screenshot displays the Astronomer web interface for a task run. On the left, a task graph shows 'index_documents' and 'publish_index' tasks, both with green checkmarks indicating success. The main panel shows details for the 'index_documents' task, which is marked as 'Success'. Below this, a table lists task metadata:

Operator	Try Number	Start Date	End Date
@task	2	2026-08-25 21:29:27	2026-08-25 21:29:43

Below the table, there are tabs for 'Logs', 'Rendered Templates', 'Storage', 'Asset Events', 'Audit Log', and 'Code'. The 'Storage' tab is selected, showing a 'Task State Store' section with a key-value pair:

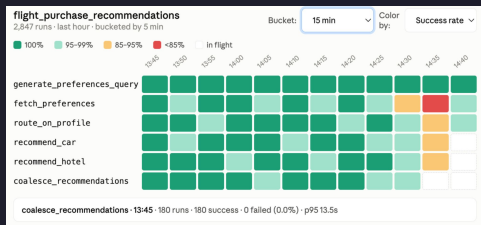
```
indexed {
  "MSL_MASTCAM_MAHLI_FOCUS_MERGE_PRODUCTS.PDF": {
    "document": "MSL_MASTCAM_MAHLI_FOCUS_MERGE_PRODUCTS.PDF",
    "attempt": 1,
    "seconds": 3.7,
    "tokens": 2986,
    "usd": 0.003142,
    "card": {
      "document_type": "Technical Specification",
      "covers_sols": "unknown",
      "summary": "MAHLI/Mastcam focus stack acquisition and merge"
    }
  },
  "MSL_MASTCAM_MARDI_INSTRUMENT_DESCRIPTIONS.PDF": {
    "document": "MSL_MASTCAM_MARDI_INSTRUMENT_DESCRIPTIONS.PDF",

```

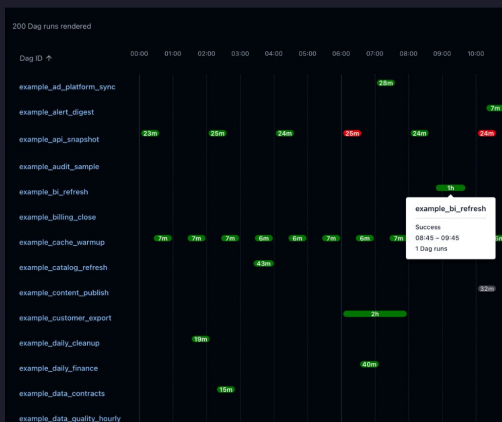
New visualizations

Minute-based Calendar View

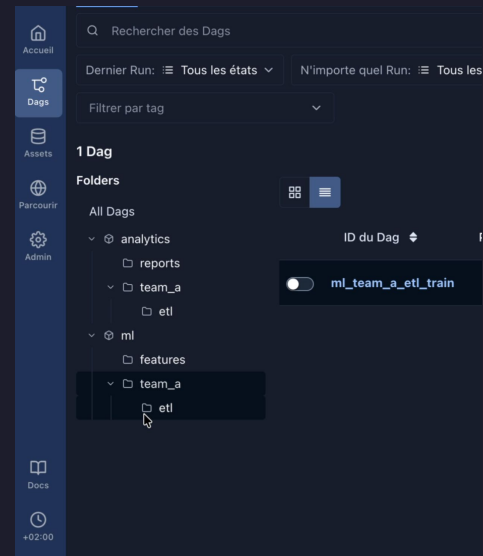
Support subsecond UI



Cross Dag Gantt Chart



Dag Folders

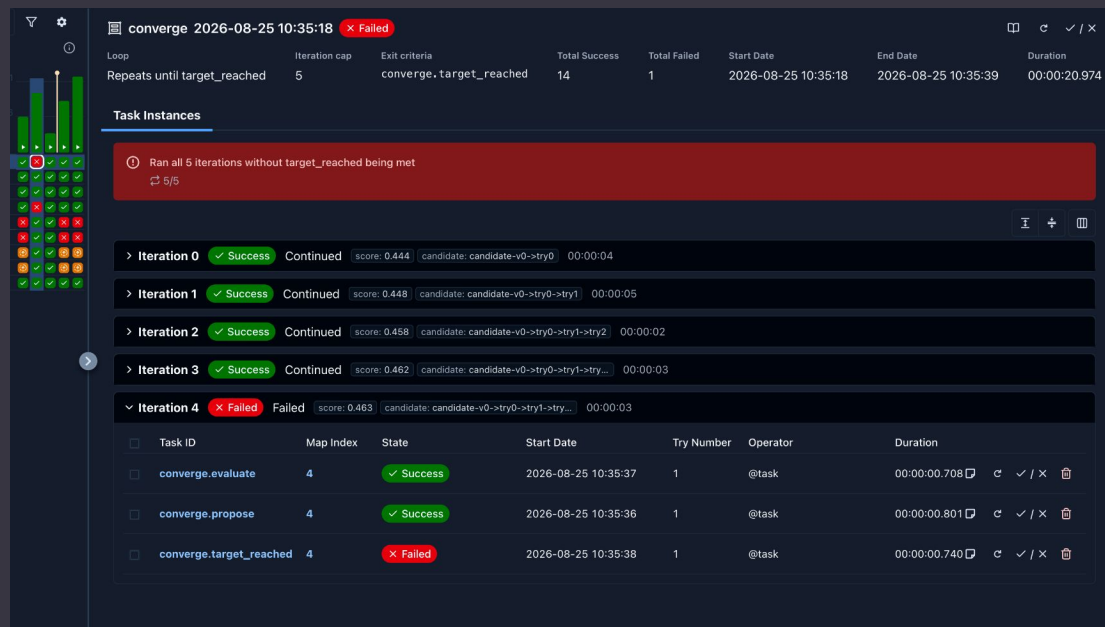


Medium Term

Next features

Task Loops

Dynamic Task Groups



Long Term

Agent UI

Do we bring Agents as a top-level concept like Dags and Assets?

Where does governance, durability, token economics and recovery get exposed in the UI?

Debugging Errors

Home Page to Task Error Logs requires navigating too much

-> Failed Task Logs Inbox like HITL

Information may be in your logs, but the root cause maybe also be elsewhere

Navigating is harder with tab bloat. For many tasks, Asset events or xcoms may be empty

-> Hide irrelevant tabs

-> New ways to plugin that isn't another tab

-> Re-architect the tab IA

Handling unstructured data

Improve UX for JSON blocks

Too many collapsed json blocks make everything hard to read

Retry Policy, Task Steps & Task Loops

All three have a “reason” which is more human readable than a dump of logs

Code defined structure

Rendered Templates and Dag Trigger
Params already exist

Integrate rendered templates into the UI instead of dumping them in a tab.

Do we let a task expose a “resource” so that we can view more than task state or duration? Tokens, evaluated AI scores,

Handling Unstructured Data

Task Loops

Expose reasons early before
diving into log files

Handling unstructured data

Improve UX for JSON blocks

Too many collapsed json blocks make everything hard to read

Retry Policy, Task Steps & Task Loops

All three have a “reason” which is more human readable than a dump of logs

Code defined structure

Rendered Templates and Dag Trigger
Params already exist

Integrate rendered templates into the UI instead of dumping them in a tab.

Do we let a task expose a “resource” so that we can view more than task state or duration? Tokens, evaluated AI scores,

Ever Growing Use Cases

Upgraded Assets UX

Use partition key mapping to connect watchers, aliases, and events with their dag runs

Dynamic Task Groups

Graph won't always be the same.

Improve Plugins

Action plugins (eg: button to open your own AI chat modal in Airflow)

More officially maintained plugins (like for common ai provider)

Plugin Registry

Discussion

These were our ideas, what are yours?

The 2026 Apache Airflow® Survey is Live!

- Takes just 7 minutes
- Get a free AI Orchestration Fundamentals Certification Exam or DAG Authoring Certification (normally \$150 each)



Content Slide - Full Width

Content Slide - Full Width with Cards

Title of Card

Text goes here

Title of Card

Text goes here

Title of Card

Text goes here

Content Slide - Full Width with Cards

Title of Card

Text goes here

Title of Card

Text goes here

Title of Card

Text goes here

Content Slide

Two Column

Content Slide

Two Column

Content Slide

Two Column

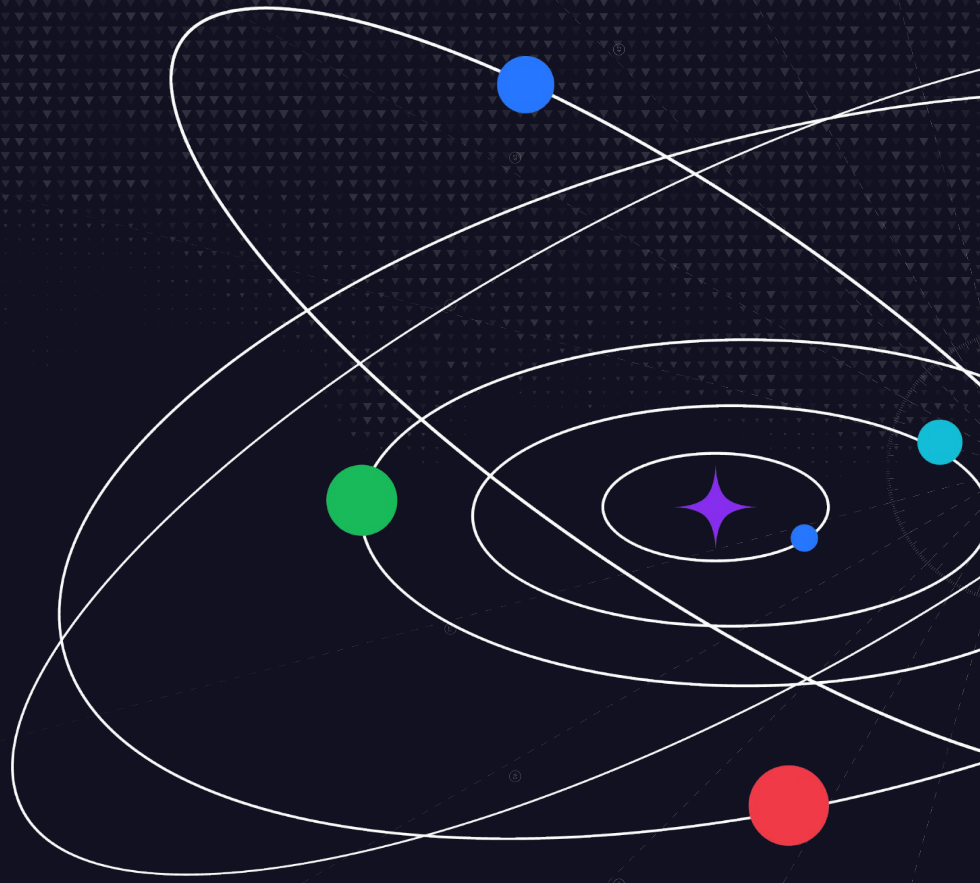
Content Slide

Two Column

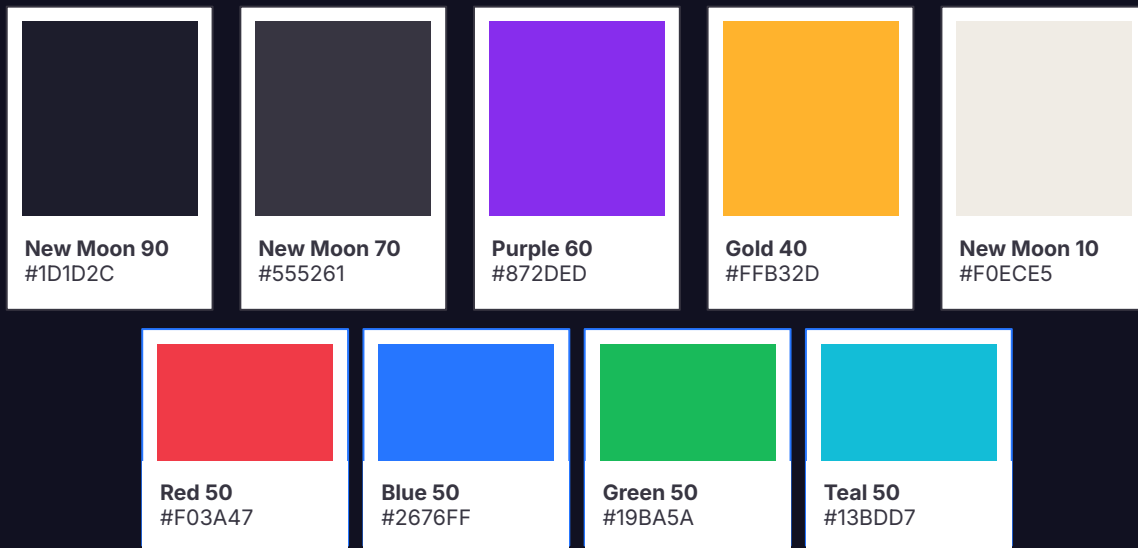
APPENDIX

For reference only - delete
before sharing

ASTRONOMER



Astronomer Brand Colors



Iconography

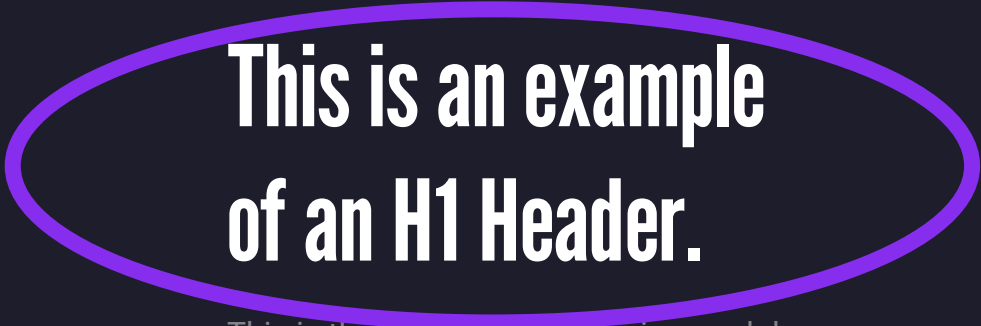


HEADERS & TITLES

League Gothic

League Gothic is a default Google Webfont that can also be [downloaded here.](#)

THIS IS AN EYEBROW



This is an example of an H1 Header.

This is the body copy. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod [tempor incididunt](#) ut labore et dolore magna aliqua. Ut ed minim veniam.

BODY & PARAGRAPHS

Roboto

Roboto is a default Google Webfont that can also be [downloaded here.](#)

THIS IS AN EYEBROW

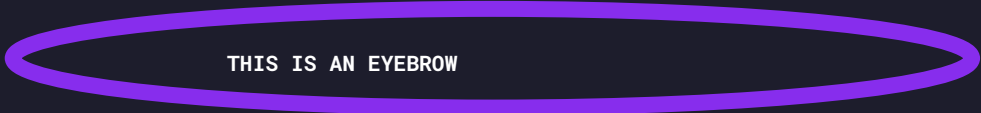
This is an example of an H1 Header.

This is the body copy. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod [tempor incididunt](#) ut labore et dolore magna aliqua. Ut ed minim veniam.

EYEBROWS & CODE

Roboto-Mono

Roboto-Mono is a default Google Webfont that can also be [downloaded here.](#)

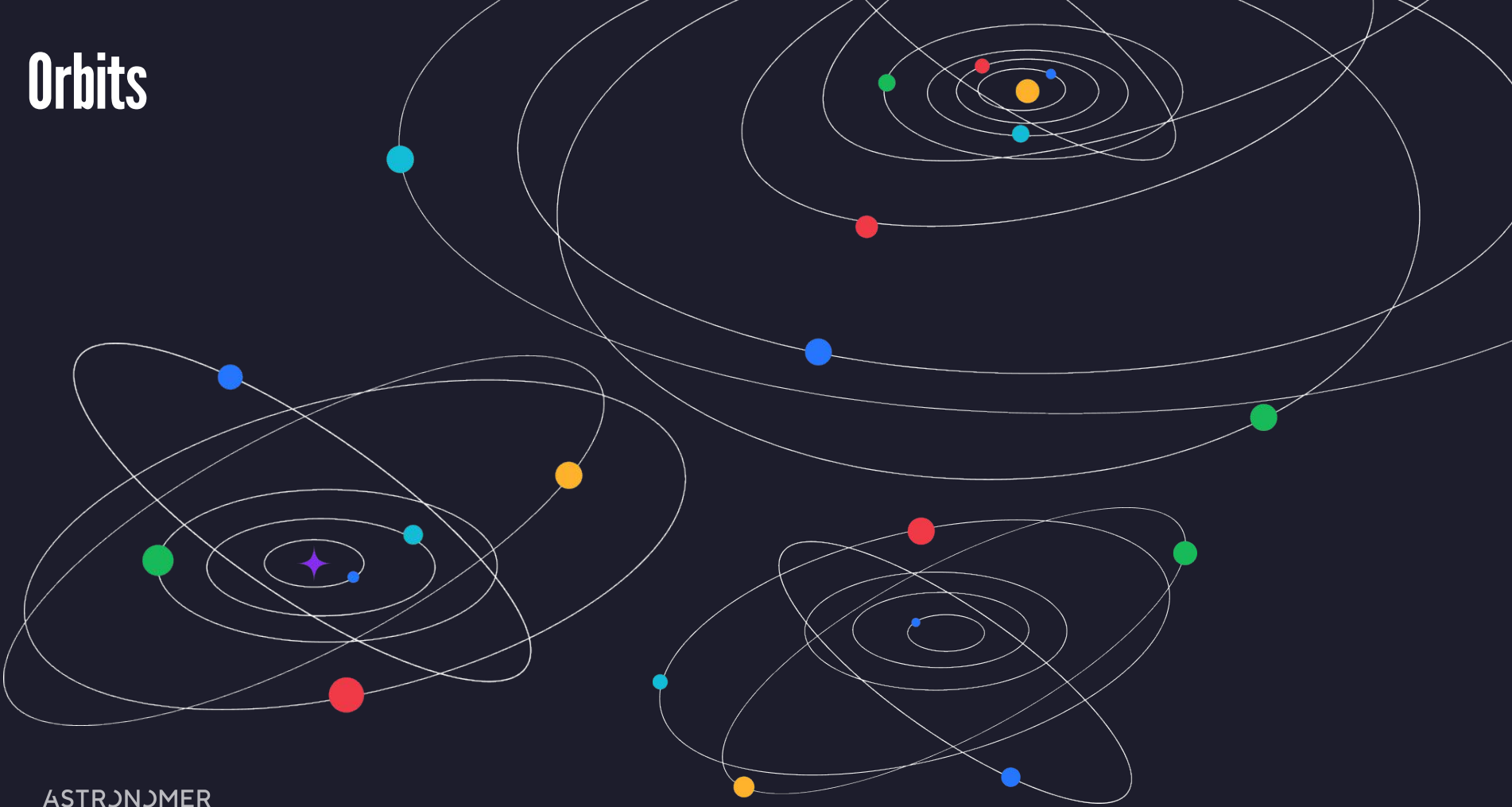


THIS IS AN EYEBROW

This is an example of an H1 Header.

This is the body copy. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod [tempor incididunt](#) ut labore et dolore magna aliqua. Ut ed minim veniam.

Orbits



ASTRONOMER

Product Illustrations

